Optical and SAR Image Fusion Based on Complementary Feature Decomposition and Visual Saliency Features

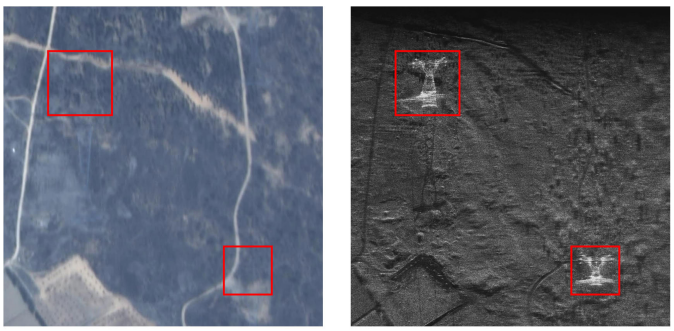
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***Abstract*— With the expansion of optical (OPT) and synthetic aperture radar (SAR) image fusion application scenarios, it is necessary to integrate their information into land classification, feature recognition, and target tracking. Current methods focus excessively on integrating multimodal feature information to enhance the information richness of the fused images, whereas neglecting the highly corrupted visual perception of the fused results by modal differences and SAR speckle noise. To address that, this article proposes a novel optical and SAR image fusion framework named visual saliency features fusion (VSFF), which is based on the extraction and balancing of significant complementary features of optical and SAR images. First, we propose a decomposition algorithm of complementary features to divide the image into main structure features (MSF) and detail texture features (DTF). Then, for the fusion of MSF, we reconstruct the visual saliency features (VSF) maps of the pixel and structure that contain significant information from optical and SAR images, and input them into a total variation constraint model to compute the fusion result and achieve the optimal information transfer. Meanwhile, we construct a new feature descriptor based on the Gabor wavelet that separates meaningful DTF from residual noise and selectively preserves features that can improve the interpretability of the fusion result. In a comparative analysis with seven state-of-the-art fusion algorithms, VSFF achieved better results in qualitative and quantitative evaluations, and our fused images have a clear and appropriate visual perception. The source code is publicly available at: https://github.com/yeyuanxin110/VSFF.**

***Index Terms*— Complementary feature, detail texture, main structure, optical (OPT) and synthetic aperture radar (SAR) image fusion, visual saliency features (VSF).**

I. Introduction

With the remote sensing image application requirement increasing and a single image offering limited information, it is necessary to integrate multimodal image data into one image to form a more abundant and meaningful fused image [1]. Recently, many researchers have focused on optical (OPT) and synthetic aperture radar (SAR) image fusion and used it to offer distinctive information for all-weather land classification, ground monitoring, and target detection [2], [3], [4]. Optical sensors passively receive information about the reflection of solar illumination from Earth objects, so that they can provide rich spectral information and sharp, detailed features that are nearly identical to those observed by the human visual, but they are vulnerable to adverse weather and insufficient illumination. In contrast, SAR is an active microwave sensor that receives backscattered energy and can acquire information under almost all weather and environmental conditions, which can capture prominent reflective targets and salient structure features at the expense of large amounts of coherent speckle noise in the image [5].



(a) (b)

Fig. 1. Set of (a) optical and (b) SAR images.

It can be seen from a set of optical and SAR images presented in Fig. 1 that there is a clear distribution between features in the optical image (e.g., roads and wasteland), but the bad weather makes the optical image have fog interference, and some of the salient targets are blurred (e.g., the power towers in the red box of the SAR image). However, the SAR sensor's extreme penetration ability and strong reflection on metal enable this target to be highlighted [5]. Therefore, fusing optical and SAR images can acquire richer target information. Moreover, due to the imaging mode of the optical sensor, two different structural objects may appear with identical spectral response information which cannot be effectively distinguished in optical imagery but can be clearly differentiated in SAR imagery [6]. As shown in Fig. 1, the spectral information

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of trees and grass in the yellow box of the optical image is similar and visually difficult to distinguish. However, their backscattered signals in the SAR images are much different and are extremely easy to distinguish. While in the red box, the boundaries of contour structures in the optical image are much less clear than those in the SAR image in areas, where the ground features are densely distributed, which is an information advantage that cannot be replaced by the SAR sensor [6]. It cannot be ignored that the random speckle noise in both sets of SAR images corrupts the feature details to varying degrees, resulting in extremely poor visual perception by the observer [7].

The complementary feature decomposition and integration of optical and SAR images is the ultimate goal of fusion. The theoretical research on complementary feature decomposition never stops, and Meyer [8] made a theoretical breakthrough by proposing to decompose the image into two types of features: structure and texture, which are regarded as bounded variation and oscillatory distribution parts in the total variation model. Based on this, there are many researchers getting inspired. Vese and Osher [9] further proposed to decompose the image into the sum of cartoon and texture parts, and then give the numerical results of the real texture image by the finite-difference method. In order to solve the problem that the numerical solution of the full variational model is computationally complex and extremely time consuming, Buades et al. [10] proposed a fast cartoon and texture image filtering algorithm, which convert Gaussian filters into nonlinear filters to achieve a simplified and fast approximation of the minimization problem of the original variational model. This contribution's property of controlling the effect of feature decomposition with only one parameter has also made it highly favored by Zhang et al. [11] and Liu et al. [12]. However, the above algorithms are mostly used to process computer vision images, and face the large amount of speckle noise in SAR images will show extremely poor feature decomposition results [7].

To design fusion rules for different feature layers of an image is always a research difficulty. Although traditional algorithms (e.g., component replacement [13] and multiscale decomposition (MSD) [14]) have excellent performance in simple image fusion tasks, they can no longer effectively integrate complementary features with great differences in the face of the increasing demand for multimodal image fusion. Many studies have attempted to solve multimodal image fusion using hybrid methods to achieve better results in complex image fusion tasks [15], [16], [17], and model-based methods [18], [19], [20] have also been used to improve the quality of generalized fusion tasks. However, it still suffers from the serious interference of SAR random noise, which causes corrupted visual perception results [6]. It is because the variational model-based approach is based on the idea of minimizing the energy function to preserve the geometric information of the SAR image by regularizing the constraints on the gradient values of the fused image [21]. However, the SAR noise is also information with high local gradient values, which is also retained in the fusion result and interferes with the fusion result.

With the wide application of deep learning [22], [23], a large number of studies have attempted to utilize the deeper feature expression capability of deep learning to fuse feature information with significant differences, which improves the information richness of the fusion results and greatly facilitates the research on advanced visual tasks [3], [4], [24]. However, the dependence of deep learning on real training datasets makes the definition of the loss function in fusion tasks often too subjective and difficult to overcome the variability in the quality of optical and SAR image data [16].

Visual saliency features (VSF) are the relatively prominent area of an image that attracts human visual attention in a bottom–up way [25]. As a basic theory in computer vision, it has been applied in remote sensing image fusion [26]. In visible and infrared image fusion, one class of algorithms typically uses a visual saliency model to compute the weights of the information retaining salient features to the fusion result [27], and another class of algorithms uses visual saliency to extract the salient regions of the source image for fusion [28]. In optical and SAR image fusion, Liu et al. [29] draw on this idea to extract SAR image information fused to optical images using a multitask saliency detection model. The above algorithms are still limited to the computation of saliency in terms of pixel grayscale criterion, which is not able to effectively integrate the abundant VSF information of multimodal images. VSF reflects the human visual behavior when freely observing images, where the observer is first attracted to areas of rich color or prominent brightness, followed by the large contours and fine edge structures, and finally by the regularly arranged textural information. It is important that the design of optical and SAR image fusion rules follow this law as much as possible to integrate important and abundant VSF information.

Therefore, we propose a novel fusion framework for optical and SAR images, named VSF fusion (VSFF). First, the complementary features of optical and SAR images are mined [10], which are divided into MSF and DTF. On the one hand, the new visual saliency maps are constructed to balance the difference features [21]; on the other hand, the effective feature descriptions are constructed to eliminate the noise interference [30]. Finally, reconstruct the fused image with real visual color [31]. Fig. 2 presents the fusion results obtained by our method and three representative fusion methods, including Laplacian pyramid (LP) [32], weighted least square (WLS) [26], and U2Fusion [33]. Obviously, our VSFF has the best visual perception and clear detail presentation while reducing a large amount of speckle noise to make the fusion result look more natural.

The main contributions of this article are as follows.

1) We propose a novel optical and SAR image fusion framework to perform effective complementary feature decomposition of source images and balanced integration of VSF and to eliminate the corruption of fusion results by SAR image noise.
2) The new pixel and structure saliency feature maps are constructed to constrain the fusion result so that it has a similar visual effect to the optical image without losing

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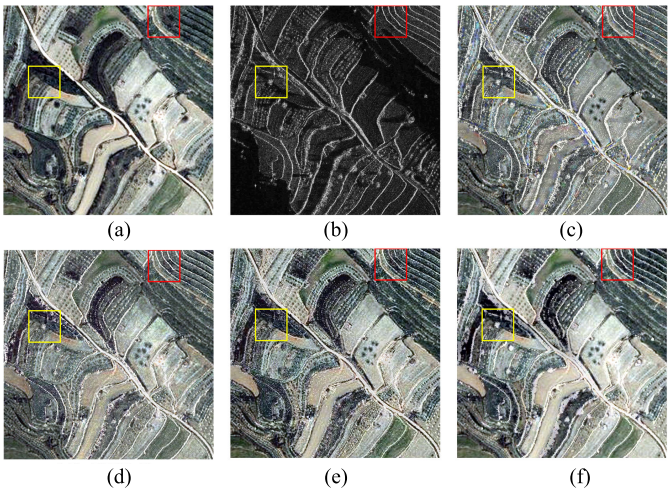


Fig. 2. Our fusion method (VSFF) is compared with some other state-of-the-art fusion methods. (a) Optical. (b) SAR. (c) LP. (d) WLS. (e) U2Fusion. (f) Our VSFF.

the significant structure and target information from the SAR image.

3) In comparison with the seven state-of-the-art fusion methods, our fusion results look more perfectly balanced with large differences in complementary feature information, improving the interpretability of the fusion results and also achieving the best results in six quantitative metrics.

II. Related Works

A. Traditional Image Fusion Method

Traditional methods have been applied and studied extensively in image fusion in the past decades by virtue of their computational simplicity and generalizability, which are divided into four main categories: component substitution methods, multiscale decomposition methods, hybrid methods, and model-based methods [6]. The representative algorithms of component substitution methods are intensity-hue-saturation (IHS) transform [34], principal component analysis (PCA) [35], Gram–Schmidt (GS) [13], and Brovery transform (BT) [36]. Although these algorithms are computationally simple, they are very dependent on the image correlation. When encountering extremely different sensor imaging results for optical and SAR images, they can result in severe spectral distortions. Among the multiscale decomposition methods, pyramid transform and wavelet transform have attracted the most attention, and the representative algorithms are Laplace pyramid [37], discrete wavelet transform (DWT) [38], curvelet [39], and non-subsampled shearlet transform (NSST) [14]. Multiscale decomposition methods overcome the drawbacks of component substitution methods, which reduce spatial and spectral distortion phenomena and provide better spatial feature localization. However, some multiscale decomposition methods trade off increased computational complexity in order to obtain better localization information. Hybrid methods combine the advantages of component substitution methods and multiscale decomposition methods to achieve better spatial and spectral information representation of fused images. The representative algorithms are IHS + DWT [40], IHS + non-subsampled contourlet transform (NSCT) [41], and modified BT [42]. The above algorithms are improved research based on the ability of component substitution methods to retain spatial details and multiscale decomposition methods to low loss of spectral information, but it is difficult to overcome the noise interference from SAR images effectively, which makes the visual effect of fused images unsatisfactory. Model-based methods are represented by variational model and sparse representation (SR) model, and the derived algorithms are variational model [43] and LP-SR [20]. The SR method treats the fusion problem as an image restoration process to generate high-resolution images by over-complete dictionary, but sparse coding and dictionary creation are quite difficult and extremely time consuming. The variational model transforms the fusion process into an energy function minimization to achieve information transfer by constructing a constraint relationship between the color information from optical images and the geometric information from SAR images.

B. Optical and SAR Fusion Method

For the information complementarity of optical and SAR image features, Kong et al. [44] have preserved the integration of optical spectral information and SAR texture information but neglected that the texture of optics and the structure of SAR also contain more interpretable features. Wu et al. [45] focus on the fusion of texture information extracted from SAR images with high-frequency detail information obtained from wavelet transform of optical images by hybrid methods, but lacks the extraction and integration of structural information. Li et al. [46] enhanced the utilization of edge features by introducing multiscale morphological gradients in the hybrid method to achieve higher correlation and lower spectral distortion of the fusion results, but still could not solve the interference of SAR image noise on visual perception. Moreover, to address the severe noise corruption problem in SAR images, Chu et al. [15] have utilized the NSST method to suppress a large amount of SAR noise to make the fusion results clearer, but the excessive suppression of noise results in the loss of many structural features of SAR. In order to avoid spectral and spatial distortion in fusion, a gain injection method [47] has been proposed to reduce the loss of information in multiscale space. Furthermore, Fu et al. [48] proposed an improved method based on phase congruence information to reduce the spectral distortion caused by nonlinear radiometric differences between images.

In contrast to the pursuit of populating more information into the fusion result, some studies have begun to focus on the direct use value of the fused image. For example, there is research [16] that focuses on improving the interpretability of SAR images by designing activity measures but without adding more meaning to the optical images. To combine significant information from the source images simultaneously, Liu et al. [29] were first proposed the saliency detection of SAR images to obtain different scales of saliency maps to enrich the information expression of optical images, but lacked a more in-depth saliency study of optical images and SAR

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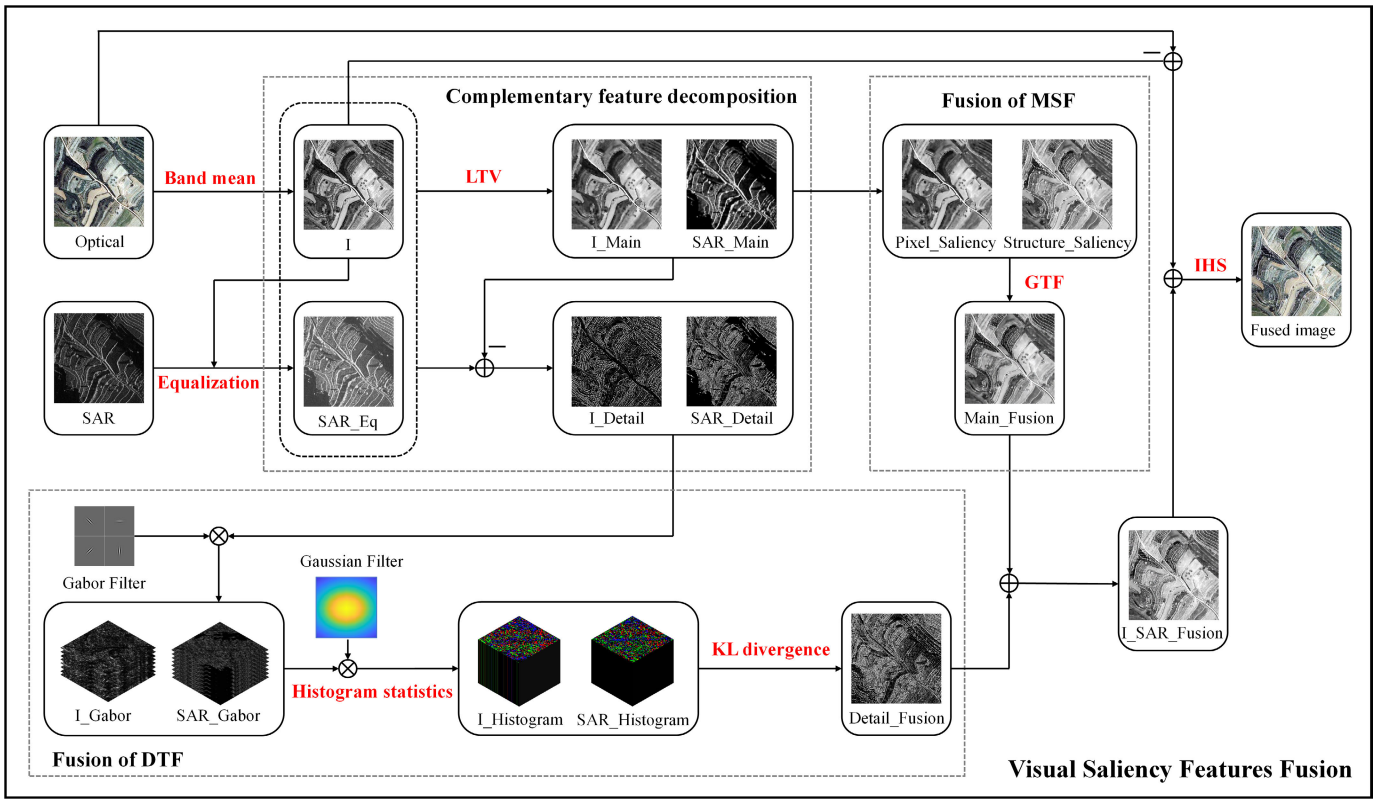


Fig. 3. Fusion framework for optical and SAR images based on VSFF.

images with multiple features. Shao et al. [19] proposed a pixel saliency-based fusion algorithm for optical and SAR images that uses a new pixel saliency map (PSM) instead of the SAR image, better preserving the spectral and spatial information but ignoring the gradient information and the destructive ability of noise in the SAR image. Zhang et al. [20] tried to design a new activity-level measurement rule to identify the extracted salient features for combining and reconstructing the fusion results, but only utilized the high-frequency information in the multiscale decomposition and ignored other important features. Gong et al. [17] propose a method based on adaptive multiscale Gaussian coprocessor filter decomposition to smooth the texture noise and retain the edge information in the texture to generate fusion results with low noise interference.

C. Deep Learning Fusion Method

Recently, deep learning has been applied to tackle scenario-specific application problems (e.g., land classification and cloud removal) for optical and SAR image fusion due to better feature extraction capabilities. An example [49] is the construction of a 3-D U-Net network combining the spectral data of optical and the texture of SAR to improve the classification accuracy of various crop types. Similarly, a self-supervised fusion framework [24] utilizing multiview contrast loss training at the super-pixel level has been proposed to improve the accuracy of land cover mapping tasks. In order to eliminate cloud contamination in the images, Grohnfeldt et al. [50] first designed a conditional generative adversarial network (GAN) to reconstruct cloud-free and haze-free optical images using auxiliary SAR images. Furthermore, Gao et al. [51] fully combine the image transformation property of convolutional neural network (CNN), and the information fusion idea of GANs to generate simulated optical images to eliminate cloudy interference regions. With the study of the complementarity between optical and SAR imagery, a paper [7] has been published proposing an unsupervised fusion network based on structure–texture decomposition, which effectively preserves the texture and enhances the structure in the fusion result. However, there is still a lack of a large amount of data with real fusion results, which makes the design of loss functions for deep learning often subjective, and the effect of fusion in different datasets varies relatively and widely [16].

In other multimodal image fusion fields, there are very advanced deep-learning fusion models have been proposed with good results. Xu et al. [33] proposed a novel unified and unsupervised end-to-end image fusion network that utilizes the adaptive information preservation degree of source images to achieve multiple fusion tasks in the same framework. Zhang et al. [52] reconstruct the fusion result by extracting the image salient features on two convolutional layers and constructing appropriate fusion rules, and the whole model achieves fully convolutional fusion training, which is a generalized fusion network.

III. Proposed Fusion Framework

In this section, we describe the VSFF framework in detail. Fig. 3 shows the whole fusion framework, which consists of four critical parts and contributions: 1) an improved complementary feature decomposition algorithm can effectively suppress speckle noise; 2) a total variation fusion algorithm that introduces VSF can enhance the overall visual perception;

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3) a novel texture feature descriptor can preserve richer detail information; and 4) a fast IHS transform fusion can supplement the realistic color information.

A. Image Preprocessing

Since the optical images used in this article are RGB multiband remote sensing images, while SAR images are single-band images, the multiband optical images need to be transformed into similar single-band data. Here, we perform a band-averaging calculation on the optical image to generate an optical intensity image

$$I = \frac{R + G + B}{3} \tag{1}$$

where I represents the optical intensity image and R, G and B are the three bands of the optical image, respectively.

Due to the large difference in spectral grayscale between SAR images and optical images, grayscale equalization is necessary before fusion to make the input images have the same grayscale mean and standard deviation to avoid severe spectral distortion

$$f_{\text{sar_Eq}} = \frac{(f_{\text{sar}} - \mu(f_{\text{sar}}))\sigma(I)}{\sigma(f_{\text{sar}})} + \mu(I) \tag{2}$$

where μ and σ represent the mean and standard deviation, respectively, and f_{sar} and $f_{\text{sar_Eq}}$ are the initial SAR image and the equalized SAR image, respectively.

B. Image Complementary Feature Decomposition

Any remote sensing image can be decomposed into a set of complementary features: MSF and DTF. The source image and the decomposed parts are defined as follows:

$$f = u + v \tag{3}$$

where f represents an optical or SAR remote sensing image, u is the MSF, and v is the DTF.

In the first step, it requires building a local indicator to divide whether each pixel belongs to MSF or DTF. MSF is the part of the image that has relatively stable local variation at different scales, while DTF is the part that tends to have large local variation after filtering. The local total variation (LTV) of the image can effectively respond to the relative degree of variation under low-pass filtering to distinguish MSF from DTF, and we define the LTV of the image and its relative reduction rate as follows:

$$\text{LTV}_\sigma(f)(x) = L_\sigma \times |\nabla f|(x) \tag{4}$$

$$\lambda(x) = \frac{\text{LTV}_\sigma(f)(x) - \text{LTV}_\sigma(L_\sigma \times f)(x)}{\text{LTV}_\sigma(f)(x)} \tag{5}$$

where L_σ is a linear filter, ∇ is the gradient calculation, λ is the relative reduction rate per pixel, and x is the pixel position of the image.

Furthermore, it is necessary to consider that there are many fragmented feature edges and speckle noise in SAR images, which will show bright spots on the image and can be easily taken as MSF. To suppress speckle noise, the Wiener wavelet is selected for smoothing the image, and this filter can adaptively adjust the filter effect based on local gray information [53].

In the second step, the decomposition of image complementary features is achieved by weighting the relative reduction rate of the original and filtered images [10]. The specific calculations for this step of the operation are as follows:

$$u(x) = w\big(\lambda(x)\big)L_\sigma \times f + \big(1 - w\big(\lambda(x)\big)\big)f \tag{6}$$

$$v(x) = f(x) - u(x) \tag{7}$$

$$w(\lambda) = \begin{cases} 0, & \lambda \le a_1 \\ (\lambda - a_1)/(a_2 - a_1), & a_1 \le \lambda \le a_2 \\ 1, & \lambda \ge a_2 \end{cases} \tag{8}$$

where w is the pixel weight calculated from λ, and a_1 and a_2 were usually taken as 0.25 and 0.5 in the experiment. We will analyze and explain the values of these two parameters taken in the experimental results in Section IV.

C. Fusion Strategy of MSF

In the fusion of MSF, structural information are presented by a combination of pixel grayscale distribution and gradient variation. Among them, the pixel grayscale distribution is the essential information to distinguish the area and type of terrestrial scene, which directly determines the overall visual effect of the fusion result. Thus, we need to extract and integrate the salient pixel grayscale information from images, such as the grayscale distribution in optical images and the high reflection value information in SAR images. While the gradient variation is an important feature information in the image that can easily attract human attention, and it is also the expression of VSF in the image. Whereas the degree of gradient variation reflects the coarse contours and fine structure of the image, it is necessary to extract saliency structure information at different scales to fuse the images. Therefore, in order to better extract VSF from images and keep visual effects [54], we reconstruct two different types of VSFM, the PSM and the structure saliency map (SSM), respectively, to constrain the fusion results.

The PSM is determined by the saliency pixels of the optical and SAR imagery. We first calculate the pixel saliency based on the degree of weight of a pixel intensity in the image among all pixels. Let I_p be the pixel intensity of pixel p in the image I. The saliency value S_p of pixel p can be defined and calculated as follows:

$$S_p = \sum_{i=0}^{L-1} M_i \big| I_p - I_i \big| \tag{9}$$

where i denotes the pixel intensity, M_i represents the total number of pixels with intensity equal to i, and $L = 256$ in the experimental images in this article. The PSM is obtained by iteratively calculating the saliency values of all pixels in the image; then, it needs to be normalized to [0, 1].

Next, u_{p1} and u_{p2} represent the PSM of optical and SAR imagery, respectively, and then, the pixel-constrained VSFM is generated by combining the pixels with higher saliency values among them

$$u_{\text{PSM}} = w u_{\text{opt}} + (1 - w) u_{\text{sar}} \tag{10}$$

$$w = \text{ceil}\big(u_{p1} - u_{p2}\big) \tag{11}$$

where u_{opt} and u_{sar} represent the MSF of optical and SAR imagery, respectively.

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Then, we generate the SSM that fuses the complementary structure features from the optical and SAR imagery. The structure features are the pixel regions of the image that are significantly expressed in the multiscale space, and the relative gradients of this part of the features are large at different scales. Also, the MSF of the terrain in the SAR image contains areas with high backscatter values. Therefore, it requires integrating pixels with large relative gradients in optical and SAR images and pixels with high-intensity values in SAR images to generate SSM that can significantly express the structure features.

The multiscale MSF image is first obtained by wiener filtering, and then, the local gradient value of the image at each scale is calculated according to the gradient operator defined as follows:

$$u^j = \text{Wiener}\left(u^{j-1}, s^{j-1}\right) \tag{12}$$

$$u_G^j = \sqrt{\left(h_x \times u^j\right)^2 + \left(h_y \times u^j\right)^2}$$
$$h_x = \begin{bmatrix} -1 & 0 & 1 \end{bmatrix} \quad h_y = h_x' \tag{13}$$

where u^j represents the j layer filtering result of MSF and s is the scale parameter of Wiener filter.

Next, the relative local gradient values can be obtained by normalizing as follows:

$$u_{\text{GN}}^j = \begin{cases} \dfrac{u_G^j}{u^j}, & 0.1 < \dfrac{u_G^j}{u^j} < 1 \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

It should be noted that the outer edges of the structure and the isolated noise points will present results greater than 1 and close to 0, respectively, and thus need to be removed. Meanwhile, an important processing step is the requirement to mean filter the relative gradient results of the images to eliminate the problem of large local gradient values due to random SAR noise. We average the relative gradient values in the scale direction to the stabilization results as follows:

$$u_{\text{GW}} = \frac{\sum_{j=1}^{n} \text{ceil}\left(u_{\text{GN2}}^j - u_{\text{GN1}}^j\right)}{n} \tag{15}$$

where n is the multiscale layers, and u_{GN1}^j and u_{GN2}^j represent the u_{GN}^j values of optical and SAR imagery, respectively.

Finally, we formulate rules to weight the multiscale relative local gradient values and pixel intensity values to generate SSM as follows:

$$u_{\text{SSM}} = (2 - k_1 \cdot k_2)u_{\text{opt}} + (k_1 \cdot k_2)u_{\text{sar}} \tag{16}$$

$$k_1 = \begin{cases} 1, & u_{GW} = 1 \\ 0.5, & \text{otherwise} \end{cases} \tag{17}$$

$$k_2 = \begin{cases} 1, & u_{\text{opt}} < u_{\text{sar}} \\ 0, & \text{otherwise.} \end{cases} \tag{18}$$

Next, the two different types of VSFMs obtained are input into the total variational model to constrain the optimization to solve the fusion results. Ma et al. [21] give a good example of a variational constraint model to optimize the results of optical and infrared image fusion named GTF. However, it only considers the information from one image, respectively, in two different constraint paradigms, which will lose more significant important information [19]. Therefore, we modify this constraint model appropriately, and the specific total variation model is constructed as follows:

$$E(x) = \frac{1}{p}\|x - u_{\text{PSM}}\|_p^p + \lambda \frac{1}{q}\|\nabla x - \nabla u_{\text{SSM}}\|_q^q \tag{19}$$

where x represents the fusion result of MSF and λ is a positive parameter, it balances the two constraints and avoids the overfitting problem.

Now, we need to consider the specific p and q norm. For the PSM constraint, we expect the best result is 0, so $p = 1$. As the gradient of the image is sparsely distributed, an approximate solution to handle the problem that $q = 0$ is NP-hard is to replace l^0 norm by l^1 norm. Thus, the gradient difference minimization problem is converted to a total variational problem. Let $y = x - u_{\text{SSM}}$, the optimization problem (19) can be rewritten as follows:

$$y^* = \arg\min_y \left\{ \sum_{i=1}^{nm} |y_i - (u_{\text{PSM}} - u_{\text{SSM}})| + \lambda J(y) \right\}$$
$$J(y) = \sum_{i=1}^{nm} |\nabla_i y| = \sum_{i=1}^{mn} \sqrt{\left(\nabla_i^h y\right)^2 + \left(\nabla_i^v y\right)^2} \tag{20}$$

where arg min represents the minimization solution and J is the first derivative of the image. Equation (20) is a standard l^1 total variation minimization problem. Rodriguez and Wohlberg [55] offer an algorithm for solving generalized total variation minimization models using the iterative weighted norm (IRN) algorithm. The algorithm can efficiently compute y and then generate the final MSF fused image.

D. Fusion Strategy of DTF

DTF is locally oscillating distribution in the image with strong repetition and orientation. Even though DTF may vary at different scales, the highly interpretable texture features in them always have a stable information representation. For that reason, a new feature descriptor is designed that can capture multidirectional and multiscale texture information within a local region of the image. It was found that the kernel function of the Gabor wavelet is similar to the human response to visual stimuli [30]. Meanwhile, the Gabor wavelet is also sensitive to the image edge information, which is an excellent texture feature filter. The following is the mathematical form of the Gabor wavelet filter:

$$g_{\lambda,\theta,\varphi,\sigma,\gamma}(x, y) = e^{-\frac{x'^2+\gamma^2 y'^2}{2\sigma^2}} \cos\left(2\pi \frac{x'}{\lambda} + \varphi\right)\gamma$$
$$x' = (x\cos\theta + y\sin\theta)$$
$$y' = (-x\cos\theta + y\sin\theta) \tag{21}$$

where λ is the wavelength parameter of the cosine function, θ is the strip direction, φ is the phase parameter of the cosine function, σ is the standard deviation of the Gaussian, γ is the spatial aspect ratio, and x and y are the coordinates inside the filter.

After obtaining the new DTF at different scales and orientations using the Gabor wavelet function, the Gaussian filtering is used to obtain the most stable DTF representation of the

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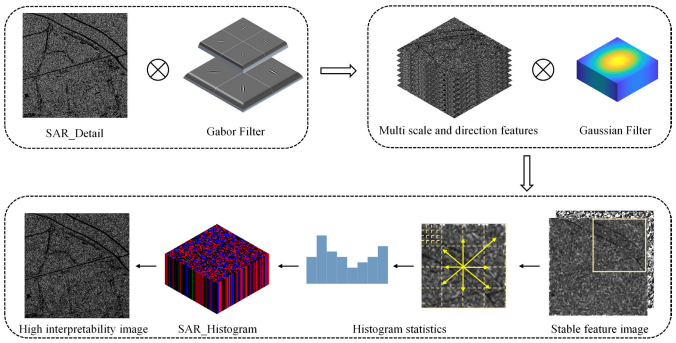


Fig. 4. Fusion process of DTF includes Gabor wavelet-based feature description and Gaussian filtering to select stable feature information, and histogram statistics to retain high interpretability features.

main scale and orientation. Subsequently, the noise that has not been eliminated needs to be further processed. Due to the fact that speckle noise appears as cluttered bright spots in the DTF, a direct feature selection operation can retain some of the noise in the fusion result. In this case, considering that the speckle noise presents an irregular distribution, while the DTF are essentially regularly gathered in a local area. Therefore, local histogram statistics can be used to improve the reliability of the DTF and eliminate the random speckle noise. After a series of operational steps, the obtained feature information has high interpretability and relative stability. The specific processing steps are shown in Fig. 4.

Finally, the high interpretability features from optical and SAR imagery are integrated, and the richer features are selected to be preserved in the fusion results. We need to compare the richness of feature information using similarity measures and local gradient values. First, the feature description vectors obtained in the previous step are normalized to form a statistical vector of feature probability distributions, and then, their similarity is measured using the KL divergence, which is defined as follows:

$$\mathrm{KL}[P\|Q] = \sum_{x\in X}\left[P(x)\log\frac{P(x)}{Q(x)}\right] \tag{22}$$

where P and Q represent the probability distribution vectors and x is the coordinates inside the vector. Specifically, KL divergence is not satisfying symmetry, i.e., $\mathrm{KL}[P\|Q] \neq \mathrm{KL}[Q\|P]$. We construct a new similarity measure by computing the average value of $\mathrm{KL}[P\|Q]$ and $\mathrm{KL}[Q\|P]$. If the value is smaller, it means that the features at that pixel are more similar. The new similarity metric value (SMV) is calculated as follows:

$$\mathrm{SMV} = \frac{\mathrm{KL}[P\|Q] + \mathrm{KL}[Q\|P]}{2}. \tag{23}$$

Next, a suitable threshold is chosen to determine whether the feature information is similar, which is defined by calculating the mean of the SMV of all pixels in the image. In summary, we give the computing steps for the fusion of DTF as follows:

$$v_f(m,n) = \begin{cases} \dfrac{v_o(m,n)+v_s(m,n)}{2}, & \mathrm{SMV}_{(m,n)} < \mathrm{SMV}_{\mathrm{mean}} \\ H(m,n), & \mathrm{SMV}_{(m,n)} \geq \mathrm{SMV}_{\mathrm{mean}} \end{cases} \tag{24}$$

$$H(m,n) = \begin{cases} v_o(m,n), & G_o(m,n) \geq G_s(m,n) \\ v_s(m,n), & G_o(m,n) < G_s(m,n) \end{cases} \tag{25}$$

where $\mathrm{SMV}_{\mathrm{mean}}$ represents the mean of the SMV of all pixels, v_o and v_s are the DTF values of optical and SAR imagery, respectively, G_o and G_s are the gradient values of DTF of optical and SAR imagery, respectively, m and n are the pixel coordinates of images, and v_f is the fusion result of DTF.

E. Fast IHS Transform Method

Since the complementary feature decomposition inevitably results in the loss of some spectral information in the image, we have to process the fused results to recover the realistic image color. The IHS fusion is a classical image fusion method, which can realize the transfer of spectral information through simple calculation [56]. It is possible to supplement optical color information to the final fusion result by replacing the original I component image with the fusion result of the optical image intensity information and SAR image in the fusion process [31]. The procedure for the fast IHS transformation method is as follows:

$$\begin{cases} R_f = R + I_f - I \\ G_f = G + I_f - I \\ B_f = B + I_f - I \end{cases} \tag{26}$$

where R_f, G_f, and B_f are the fusion result of RGB bands, R, G, and B are the RGB bands of optical image, and I_f is the fusion results of the intensity components.

IV. Experimental Results

In this section, the proposed method is compared with seven state-of-the-art fusion methods: LP [32], dual-tree complex wavelet transform (DTCWT) [57], NSCT [58], hybrid-MSD [59], WLS [26], image fusion CNN (IFCNN) [52], and U2Fusion [33]. Among them, the first three are classical multiscale decomposition methods, the fourth is the latest hybrid multiscale decomposition method, the fifth is a fusion method based on visual saliency maps, and the last two are the latest advanced image fusion network in deep learning.

In our experiments, the quality of image fusion is evaluated in qualitative and quantitative terms. Qualitative evaluation is a visual perception analysis of the overall image and local details of the fusion result. There are some differences in the visual perceptual focus of different source images. A large number of index theories have emerged in image fusion for quantitative evaluation [60], including the measurement of image characteristics such as image information, gradient, and structural similarity. Each of the image evaluation indexes has its advantages and disadvantages, it is necessary to synthesize multiple indexes. Six evaluation indexes: entropy (EN) [61], mutual information (MI) [62], Qabf [63], Qo [64], visual information fidelity fusion (VIFF) [65], and sum of correlation of differences (SCD) [66] are adopted in this article. EN is an objective evaluation index that measures the information contained in an image. MI is a measure of the degree of similarity between two images (i.e., how much information of the original image is contained in the fusion result). Qabf

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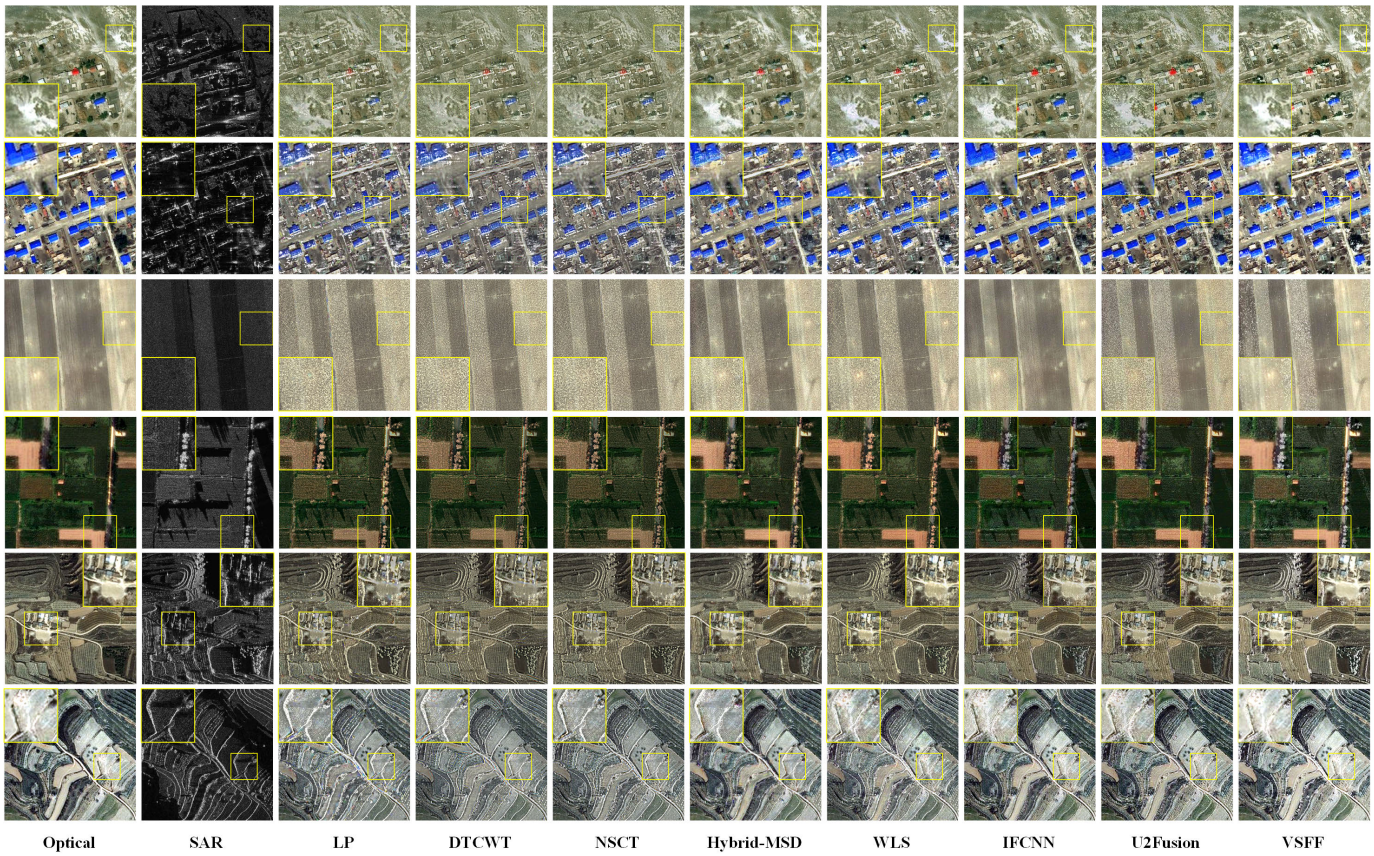


Fig. 5. Qualitative evaluation of fusion results from six different scenarios (from the high-resolution SAR and optical dataset). From left to right are optical and SAR images, LP, DTCWT, NSCT, hybrid-MSD, WLS, IFCNN, U2Fusion, and our VSFF fusion results.

is a measure of how much significant information of the input image is represented in the fusion result. Qo is a novel generalized image quality index. VIFF is a proposed measure of the generalized image quality based on the fidelity of visual information. SCD is a novel evaluation index based on the sum of correlation coefficients between the difference image and the original image. The above-mentioned metrics' higher values mean better results.

A. Datasets and Parameter Settings

To verify the fusion effect on optical and SAR salient features and noise removal capability of the experimental algorithm. All the algorithms are tested on a high-resolution (submeter level) SAR and optical dataset and a publicly available WHU-OPT-SAR dataset. Here is a detailed description of the datasets.

1) High-Resolution SAR and Optical Dataset: The dataset [67] includes the high-resolution SAR images (0.5 m) acquired from the surrounding areas of Baicheng City in Jilin Province and Weinan City in Shaanxi Province, including three typical ground object scenes of houses, farmland, and mountains.[1] After downloading the Google-Earth optical images of the corresponding areas and performing high-precision image registration by the methods in CFOG [68], [69], [70], they were further cropped into 1000×1000 pixel image pairs, and then 60 pairs of images with abundant information of scenes were selected as the test dataset.

2) WHU-OPT-SAR Dataset: Li et al. [71] open sources a set of optical and SAR image datasets collected in Hubei province. Optical images are from GF-1 satellite (2-m resolution), and SAR images are from GF-3 satellite (5-m resolution). WHU-OPT-SAR dataset covers a wide range of areas, including diverse terrains such as mountains, woodlands, hills, plains, and vegetation. For better showing the fusion details, the image of the dataset was resampled into the same ground sampling interval (i.e., 5 m), and was cropped to the size of 1000×1000 pixels in the experiment.

3) Parameter Settings: In the complementary feature decomposition module, the size of the Wiener filter is set to 3. In the MSF fusion module, the positive parameter λ is set to 2. In the DTF fusion module, the scale of the Gabor wavelet is set to [4, 8], the direction is set to [0°, 45°, 90°, 135°]. Other parameters are defaulted.

B. Result Analysis of High-Resolution SAR and Optical Dataset

The fusion results are shown in Fig. 5, we have enlarged some fusion details with a yellow rectangular box. And, we show the results for six of the scenes, which cover typical feature types such as houses, farmland, and mountains. In general, all the methods have a good fusion of the house, road, and field contour boundaries in the SAR images, as well as the SAR unique information such as tree shadows and

[1]The dataset is available at https://github.com/yeyuanxin110/YYX-OPT-SAR.

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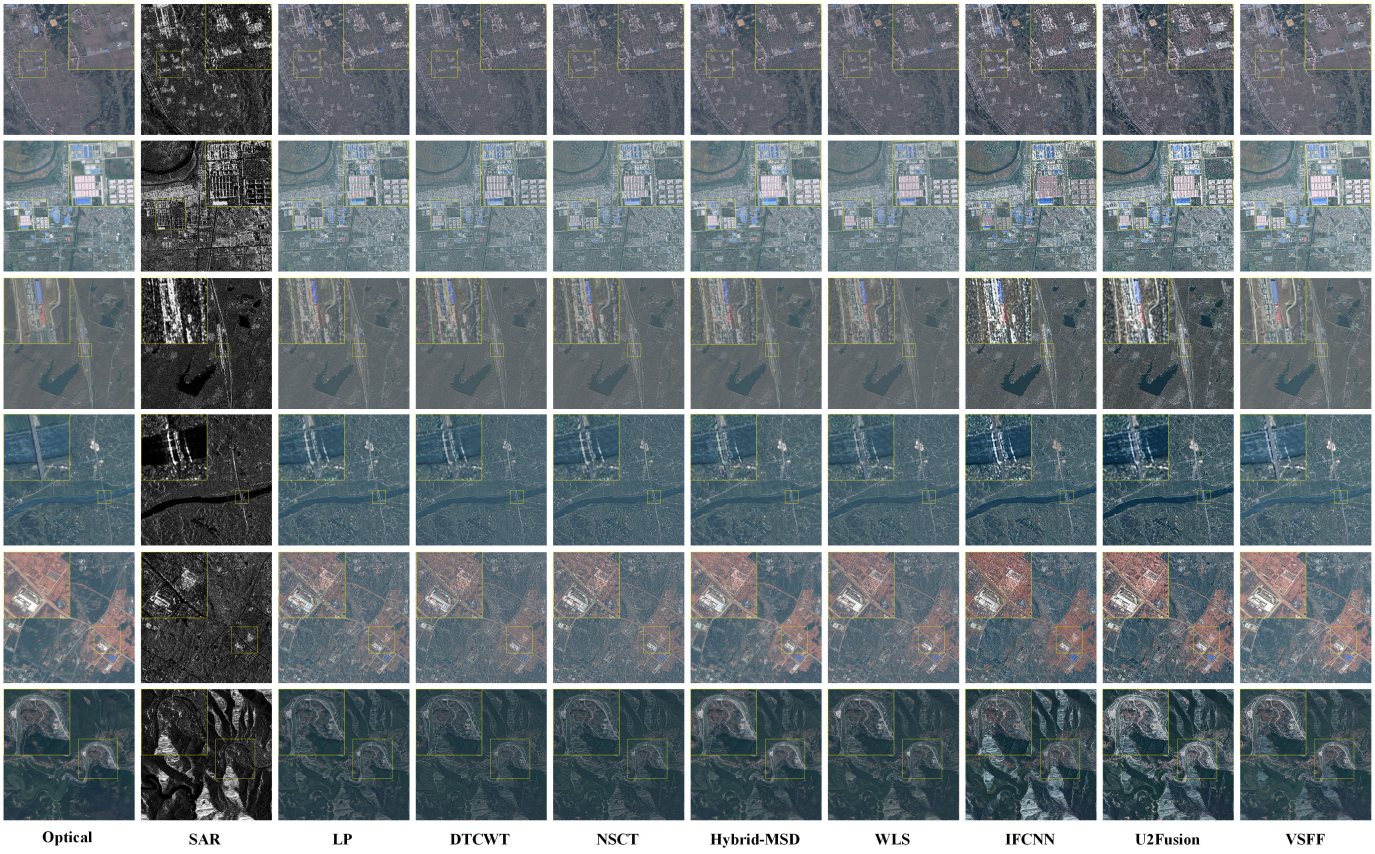


Fig. 6. Qualitative evaluation of fusion results from six different scenarios (from the WHU-OPT-SAR dataset). From left to right are optical and SAR images, LP, DTCWT, NSCT, hybrid-MSD, WLS, IFCNN, U2Fusion, and our VSFF fusion results.

field crop textures. However, in the three classical multiscale decomposition methods, LP, DTCWT, and NSCT, the spectral information of optical images is seriously damaged. This is due to the low gray-level information in the SAR image, resulting in an overall darkening of the fusion result, and some important ground object scenes are covered by shadows. In contrast, while hybrid-MSD retains better optical color information, it lacks some key salient information. For example, in the closed-up scene in the fifth row, it can be seen that hybrid-MSD is missing the field edge information provided by SAR in the lower left. Similarly, in the closed-up scene in the sixth row, the SAR tree shadow interference causes it to lose the important field path information provided by the optical image. The overall vision of the WLS fusion results is still disturbed by the hue and noise of the SAR images, and there is some spectral distortion. In the deep learning approach, the fusion results of U2Fusion are severely corrupted by SAR image noise; for example, in the first-, third-, and sixth-row scenes, the noise has obscured the feature information. While IFCNN reduces the visual impact of noise, it does not really eliminate the speckle noise, for example, in the fourth-row scene, which still leads to spectral distortion. From all closed-up views, the VSFF method achieves the best optical color fusion and highlights the visual perception of SAR main contour information while eliminating SAR image noise interference. Benefiting from the advantage of the selection of salient features, the unique details of each optical and SAR are perfectly preserved without interfering with each other.

TABLE I

QUANTITATIVE EVALUATION OF FUSION RESULTS (FROM THE HIGH-RESOLUTION SAR AND OPTICAL DATASET), WITH THE BEST RESULTS HIGHLIGHTED IN RED AND THE SECOND BEST RESULTS IN BLUE

Methods	EN	MI	Qabf	Qo	VIFF	SCD
LP	7.206	1.218	0.470	0.384	**0.571**	1.650
DTCWT	7.119	1.118	0.413	0.360	0.480	1.603
NSCT	7.151	1.192	0.459	0.393	0.523	1.652
Hybrid-MSD	**7.329**	1.281	0.451	0.402	0.474	1.724
WLS	7.214	1.458	0.450	0.419	0.502	**1.750**
IFCNN	7.219	1.600	0.436	0.404	0.475	1.625
U2Fusion	7.264	**1.923**	**0.536**	**0.424**	0.498	1.734
VSFF	**7.383**	**3.619**	**0.604**	**0.474**	**0.603**	**1.813**

Fig. 7 and Table I show the quantitative analysis of the fusion results from different methods on the high-resolution SAR and optical dataset. We can see that VSFF outperforms other methods in all metrics, which indicates that VSFF has a great advantage in integrating structure and detail information.

C. Result Analysis of WHU-OPT-SAR Dataset

The fusion results are shown in Fig. 6, specifically containing important targets such as railroads, bridges, and wastelands. The VSFF fusion results have a more detailed representation of the salient feature information and excellent denoising effect for different images. From the perspective of visual perception of fusion results, our method demonstrates excellent denoising effects in all scenes. On the one hand,

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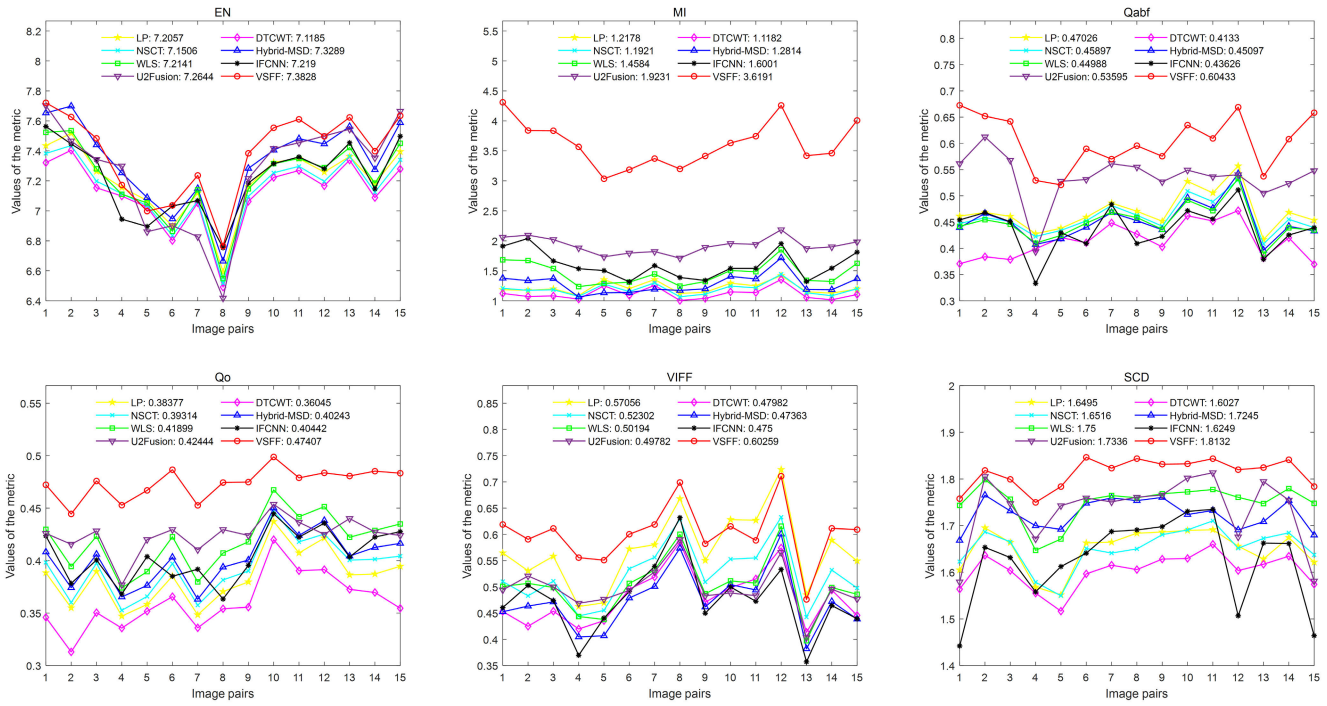


Fig. 7. Quantitative analysis of fusion results (from the high-resolution SAR and optical dataset).

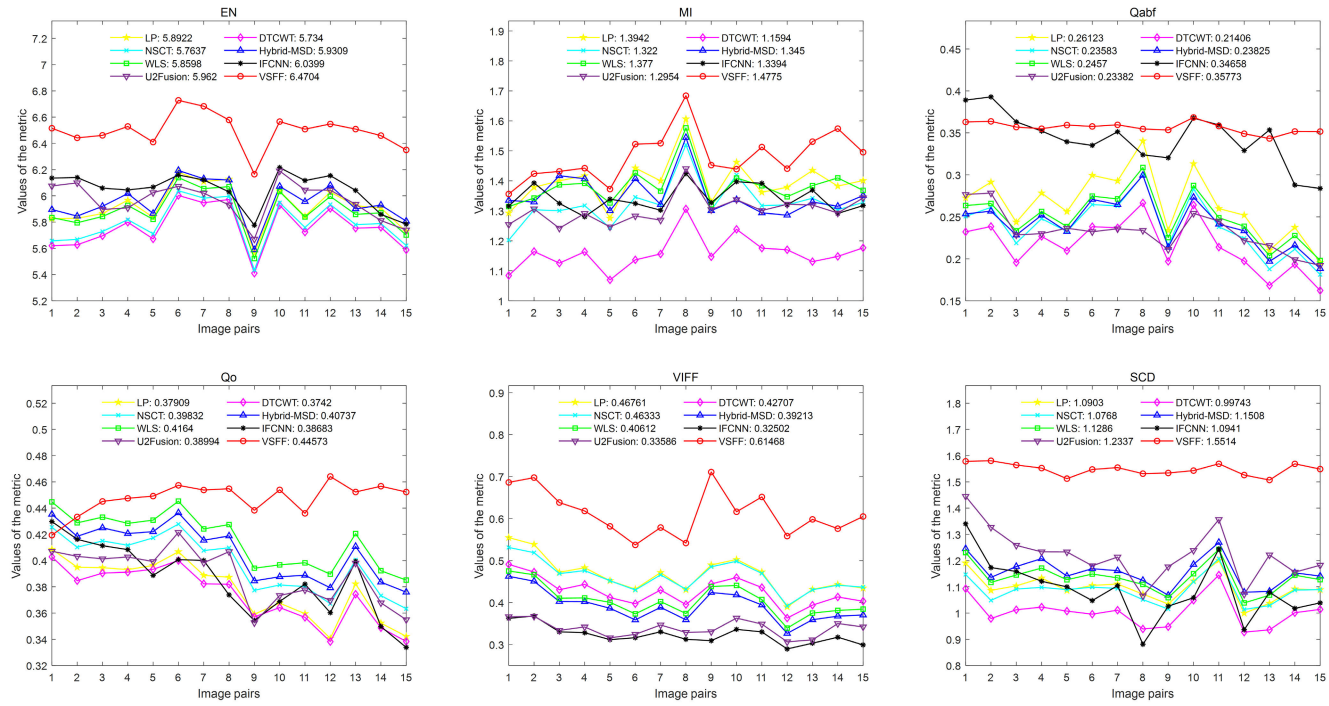


Fig. 8. Quantitative analysis of fusion results (from the WHU-OPT-SAR dataset).

the closed-up areas of the fusion result map look cleaner and more comfortable, and the optical and SAR features are clearly distinguished. Especially in the first, second, and fifth scenes, the VSFF method cares more about the real texture and color information of the land and houses than other methods and avoids having them obscured by the cluttered speckle noise. On the other hand, the VSFF method can accurately retain salient and important target information. For example, in the third and fourth scenes, the red trains and bridges in the optical images are highly valuable features. However, it is obvious that the fusion results of other methods show blurred or even nonexistent, only our method reconstructs the target completely. In conclusion, the VSFF fusion method achieves effective removal of SAR image noise without losing important feature information and restores the real color of ground objects to the maximum extent.

Fig. 8 and Table II show the quantitative analysis of the fusion results from different methods on the WHU-OPT-SAR

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TABLE II
QUANTITATIVE EVALUATION OF FUSION RESULTS (FROM THE WHU-OPT-SAR DATASET), WITH THE BEST RESULTS HIGHLIGHTED IN RED AND THE SECOND BEST RESULTS IN BLUE

Methods	EN	MI	Qabf	Qo	VIFF	SCD
LP	5.892	**1.394**	0.261	0.379	**0.468**	1.090
DTCWT	5.734	1.159	0.214	0.374	0.427	0.997
NSCT	5.764	1.322	0.236	0.398	0.463	1.077
Hybrid-MSD	5.931	1.345	0.238	0.407	0.392	1.151
WLS	5.860	1.377	0.246	**0.416**	0.406	1.129
IFCNN	**6.040**	1.339	**0.347**	0.387	0.325	1.094
U2Fusion	5.962	1.295	0.234	0.390	0.336	**1.234**
VSFF	**6.470**	**1.478**	**0.358**	**0.446**	**0.615**	**1.551**

dataset. Among them, the EN and MI indexes show the high information content of the VSFF fusion results, which intuitively indicates that the salient features extracted are the high interpretability features of the images. In particular, the higher VIFF index indicates that the fusion results have better clarity. Thus, the fusion results of VSFF are more suitable for visual interpretation.

D. Experimental Analysis and Ablation Study

1) Complementary Feature Decomposition Based on Wiener Filter: To verify the excellent performance of the proposed Wiener filter-based complementary feature decomposition in feature preservation and SAR noise removal, a comparative analysis of SAR image feature decomposition results with other filters is performed. We selected the most classical Gaussian filter and RGF [72], which have been widely used in a large number of complementary feature decomposition experiments, and the scale parameters of all filters in the experiments were identical.

Fig. 9 shows the MSF results of complementary feature decomposition of SAR images by different filters. It can be seen that the result of the Gaussian filter eliminates the noise but also blurs the structural feature information, which is due to the fact that the Gaussian kernel treats all pixels in the image equally. And, the results of RGF highlight the structural feature information well. But as an edge-preserving filter, it cannot distinguish well between noise and structure and thus preserves the noise almost entirely as well. The Wiener filter adaptively adjusts the filtering effect to achieve the best results in structure preservation and noise suppression at the same time, and the MSF image looks clear and clean. Table III presents the comparison of quantitative metrics of the fusion results using different filters, and the Wiener filter-based results are the best.

2) Different Types of VSF Method: We performed an experimental analysis of the effectiveness of the proposed two types of VSFM. Fig. 10 presents the fusion results for the four cases. When no VSFM is input, the fusion results are completely corrupted by the noise of the SAR image, and even severely blurred resulting in very poor visual perception. When only one of the VSFMs is input, the blurring of the image improves, but spectral distortion or loss of structure occurs. Only when the complete PSM and SSM are input, the fusion result perfectly balances the difference feature information and has the best visual effect. Table IV presents the quantitative evaluation of different fusion results in the four cases. The metrics results of the first three tests have their own advantages, while the complete VSFM achieves the best results in all metrics.

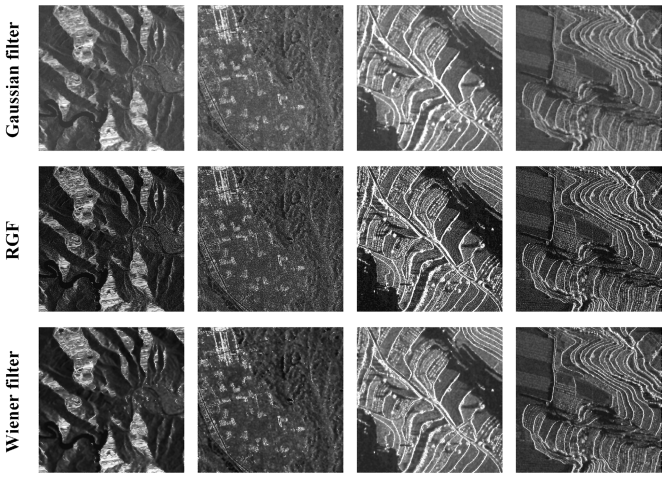


Fig. 9. MSF results of complementary feature decomposition of different filters, from top to bottom, are Gaussian filter, RGF, and Wiener filter.

TABLE III
QUANTITATIVE EVALUATION OF FUSION RESULTS FOR DIFFERENT FILTERS, WITH THE BEST RESULTS HIGHLIGHTED IN BOLD

	EN	MI	Qabf	Qo	VIFF	SCD
Gaussian	6.756	1.916	0.231	0.381	0.331	1.299
RGF	6.792	2.040	0.262	0.413	0.402	1.362
Wiener	**6.812**	**2.281**	**0.450**	**0.457**	**0.610**	**1.649**

3) Analysis of Noise Removal Capability of DTF Fusion Strategy: In the DTF fusion processing, we further eliminate the noisy information that is considered as DTF. As can be seen from Fig. 11, our algorithm can effectively remove a large amount of speckle noise, so that the real texture information of the image is clearly displayed. This is due to the multiscale and directional information extraction and representation of the initial DTF using Gabor descriptors in order to accurately segment the random noise and the regularly recurring texture. Table V presents the quantitative evaluation of the fusion results with and without noises, and our noise removal algorithm has higher metric values.

4) Parametric Analysis of Complementary Feature Decomposition: Fig. 12 presents the variation of λ for pixels in different regions for different sizes of convolution kernels. In the detail texture region, λ increases rapidly with the size of the convolution kernel, approaching 1. In the structure edge region, it varies slowly with the size of the convolution kernel, below 0.25. The two key parameters in (8) are a_1 and a_2, which are used to determine the structural or textural attribute weights of the pixels. In order to ensure the feature decomposition effect of optical and SAR images, the size of the convolution kernel of the filter is generally between 3 and 7, so the λ values of pixels with texture attributes are all greater than 0.5, while those of pixels with structural attributes are all less than

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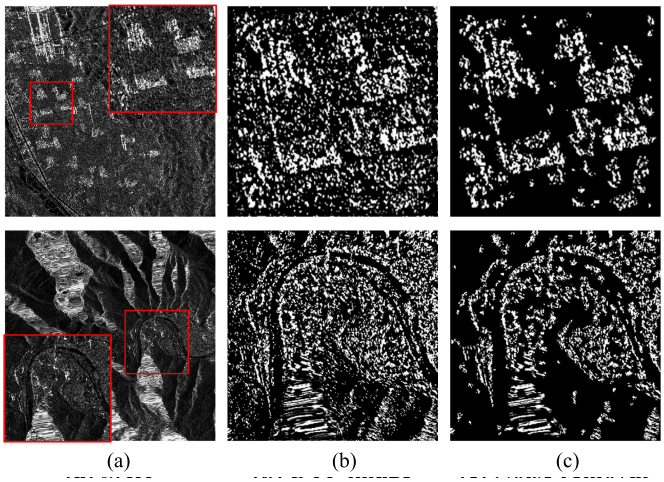
(a) (b) (c) (d) (e) (f)

Fig. 10. Comparative analysis of VSFM method. (a) Optical. (b) SAR. (c) Without VSFM. (d) Only PSM. (e) Only SSM. (f) PSM + SSM.

TABLE IV

QUANTITATIVE EVALUATION OF FUSION RESULTS FOR DIFFERENT TYPES OF VSFM METHOD, WITH THE BEST RESULTS HIGHLIGHTED IN BOLD

	EN	MI	Qabf	Qo	VIFF	SCD
Without VSFM	6.639	1.487	0.293	0.292	0.215	0.885
Only PSM	6.713	1.740	0.299	0.293	0.238	1.110
Only SSM	6.543	2.032	0.282	0.427	0.471	1.102
PSM+SSM	**6.812**	**2.281**	**0.450**	**0.457**	**0.610**	**1.649**



(a) (b) (c)

Fig. 11. Comparison of DTF image with noise and noise removal. (a) SAR. (b) DTF image. (c) Noise removal.

TABLE V

QUANTITATIVE EVALUATION OF FUSION RESULTS WITH AND WITHOUT NOISES, WITH THE BEST RESULTS HIGHLIGHTED IN BOLD

	EN	MI	Qabf	Qo	VIFF	SCD
With noise	6.801	2.149	0.383	0.429	0.560	1.586
Without noise	**6.812**	**2.281**	**0.450**	**0.457**	**0.610**	**1.649**

0.25. Therefore, $a_1 = 0.25$ and $a_2 = 0.5$ are a reasonable choice of parameter values.

5) Runtime Analysis of Fusion Methods: Table VI presents the average runtimes of different fusion methods tested on the datasets, which were all run on computers with Intel Core i7-11800H CPU. Our method is much faster than the traditional multiscale method NSCT and saves more time even compared with the latest hybrid-MSD and WLS methods. However, it runs slower than other methods, which is mainly due to the iterative optimization in fusion and the statistical

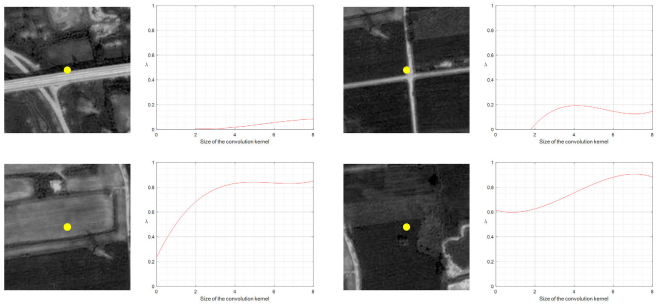


Fig. 12. Variation of the value of λ for pixels in different regions for different sizes of convolution kernels. (Top) Structure edge pixels. (Bottom) Detail texture pixels.

TABLE VI

AVERAGE RUNNING TIME (IN SECONDS) OF DIFFERENT FUSION METHODS TESTED ON THE DATASETS

Methods	Running time	Methods	Running time
LP	0.0485	WLS	12.5946
DTCWT	1.5862	IFCNN	3.1773
NSCT	51.6623	U2Fusion	4.5861
Hybrid-MSD	10.8275	VSFF	7.8032

computation in noise removal. Moreover, our algorithm was written and implemented in MATLAB, which lacks sufficient code optimization in some matrix operations.

V. CONCLUSION

In this article, we propose a novel optical and SAR image fusion framework named VSFF based on VSF. It extracts the salient and complementary information of the image and then achieves the purpose in different fusion methods and rules. From the fusion results, it is obvious that our method eliminates more noise and retains the salient and important feature targets in both optical and SAR images. Surely, our method achieves the best results on six different quantitative evaluation metrics compared with the seven state-of-the-art fusion methods. It proves that our results have abundant spectral information and clearer visual perception. However, although our algorithm can handle a large number of optical and SAR images well when faced with poor image quality due to bad conditions such as haze and heavy clouds in optical images, it is difficult to completely eliminate the effect of cloud damage on structures due to the limitations of VSF extraction, which degrades the quality of the fusion results.

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We will improve and optimize it in our future work so that it can be widely adapted to optical and SAR images under different conditions.

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